

SCORING, SUBSTITUTION MATRICES AND SCORE AGGREGATION FOR TCR ANTIGEN-SPECIFICITY SEARCH

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vdjmatch technical appendix — draft, regenerated from `bench/gen_vdjam.py`, `bench/loo_vdjam.py`,
`bench/gen_appendix_assets.py`

Abstract

`vdjmatch` annotates T-cell receptor (TCR) clonotypes by antigen specificity through fuzzy CDR3 search against a labelled database (VDJdb), built on the `seqtree` search core. This note documents the *scoring* layer: how CDR3 similarity is scored, the data-driven TCR substitution matrix (VDJAM) and why it is region-structured, and how per-hit scores are aggregated into a control-calibrated significance. CDR3 loops resist the assumptions behind BLOSUM/PAM — they cannot be multiply aligned across clonotypes of different length, and germline-encoded flanks (CASS...) are so over-conserved that a naive matrix conflates germline bias with selection. We give a position-debiased log-odds estimator that isolates the substitution signal, show empirically that this signal lives in the non-template (NDN) core while the V/J flanks are germline-determined, and demonstrate the consequences on worked examples (GIL, NLV) using the alignments and CIGAR strings produced by `seqtree`. The E-value theory used for significance is derived in the companion `seqtree` appendix; here we summarize it and give the paired-chain extension.

1. THE SEARCH-AND-SCORE FRAMEWORK

Given a query CDR3 q and a labelled reference set D (VDJdb), `vdjmatch` retrieves the neighbours of q within a fixed scope $\theta = (s, i, d, t)$ (max substitutions, insertions, deletions, total edits) using the `seqtree` engine, which defines a ball $B_\theta(q) = \{x : \text{edit}(q, x) \leq \theta\}$ and returns, per hit, the edit decomposition and a matrix-weighted alignment. The matrix enters as a *ranking* score inside the ball; the ball itself (an edit budget) bounds the candidate set. Each hit carries its antigen labels, an extended CIGAR (`=/X/I/D`), and an alignment score. Significance comes from comparing the neighbour count to a matched background control (§3).

2. SUBSTITUTION SCORING AND THE VDJAM MATRIX

HOW BLOSUM IS BUILT, AND WHY IT DOES NOT TRANSFER.. BLOSUM is a log-odds matrix estimated from the BLOCKS database of *ungapped multiple-alignment blocks* of conserved protein regions. Its recipe is: (i) take many trusted blocks — columns of homologous, gaplessly aligned

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residues; (ii) cluster sequences within a block above a percent-identity threshold and down-weight within-cluster pairs so abundant near-duplicates do not dominate; (iii) count, over all blocks, how often residues a and b co-occur in the *same alignment column*, q_{ab} ; (iv) score $s(a, b) = \frac{1}{\lambda} \log(q_{ab}/p_a p_b)$ against the *global* background frequencies p_a . The signal is “which residues substitute for one another within conserved, column-homologous positions,” and the background is position-independent.

CDR3 loops break every step. (i) There is no trusted multiple alignment: same-specificity CDR3s differ in length and have no column homology, so “alignment columns” do not exist. (ii) The germline-encoded ends are over-conserved — essentially every human TRB CDR3 begins CASS and ends in a short J motif — so a *global* background p_a is wrong: an alanine in the N-terminal CASS is germline-fixed and carries no specificity signal, while an alanine in the centre does, yet BLOSUM would score them identically. A single global matrix estimated over all positions thus conflates germline conservation with antigen-driven selection.

WHAT VDJAM CHANGES (POINT BY POINT).. VDJAM keeps the log-odds idea but replaces the two broken ingredients. (a) *Observation unit*: instead of columns of a multiple alignment, the unit is a *same-antigen single-substitution pair* — two CDR3s annotated to the same epitope, equal length, differing at exactly one position; this is the only “aligned residue pair” obtainable without a multiple alignment, and (like a BLOCKS column pair) it records two residues that are interchangeable without changing function (here, specificity). (b) *Background*: instead of a global p_a , a *position-specific* background $\text{bg}(\cdot | L, p)$ (the residue frequency at that length and position) is used, so over-conserved columns are normalized away analytically rather than masked by hand. (c) *Redundancy*: BLOSUM’s percent-identity clustering is replaced — partly by the position-specific background (germline-shared residues are “expected” and contribute little), and prospectively by the control-calibrated significance layer (§3) that already discounts convergent public clones; per-clonotype clustering can be added exactly as in BLOSUM if needed. (d) *Scope*: BLOSUM is one global matrix; VDJAM is estimated per chain and per region, because (next paragraph) only the NDN core carries a transferable substitution signal.

OBSERVATION UNIT AND ESTIMATOR.. We replace “alignment columns” with *same-antigen single-substitution pairs*: ordered pairs of CDR3s annotated to the same epitope, of equal length, differing at exactly one position — the only clean substitution signal extractable without a multiple alignment. For a substitution $a \leftrightarrow b$ observed at position p in a CDR3 of length L on chain c , let $\text{bg}(\cdot | L, p)$ be the residue frequency at position p among all length- L chain- c CDR3s. With $f(a, b)$ the observed count of $\{a, b\}$ events and

$$e(a, b) = \sum_{\text{event slots } (L, p)} n_{L, p} [\text{bg}(a | L, p) \text{bg}(b | L, p) + \text{bg}(b | L, p) \text{bg}(a | L, p)], \quad \text{VDJAM}(a, b) = \log_2 \frac{f(a, b) + \alpha}{e(a, b) + \alpha}, \quad (1)$$

where $n_{L, p}$ is the number of observed events at (L, p) and α a Laplace pseudocount. The key point is that the background is *position-specific*: at an over-conserved column the residue frequency is peaked, so substitutions there are both rare (small f) and “expected” (matched e), and contribute ≈ 0 to the log-odds — the germline bias is removed analytically rather than by hand-masking. The matrix is consumed by `seqtree` via the squared-distance penalty $\text{pen}(a, b) = s_{aa} + s_{bb} - 2s_{ab}$ (`SubstitutionMatrix.from_similarity`); a dominant uniform diagonal makes higher learned sim-

ilarity rank a substitution better.

REGION STRUCTURE: NDN IS LEARNED, V/J ARE GERMLINE-DETERMINED.. A CDR3 is V-germline prefix · NDN insert · J-germline suffix. Estimating (1) separately by region shows (Fig. 1) that the substitution signal — the agreement with general amino-acid chemistry, measured as correlation with BLOSUM62 — is concentrated in the NDN core (TRB $r = 0.42$, TRA 0.35), whereas the V flank carries almost none (TRB 0.08, TRA 0.07): V-flank “substitutions” are dominated by germline allele differences, not free chemistry. This motivates a two-part model: **VDJAM is learned for the NDN core**, while the **V and J flanks are auto-computed** from the germline V/J sequences and their trimming. Concretely, a CDR3 position derived from V- (or J-) germline is near-invariant when it survives trimming and reverts to the random-insert background when trimmed; so the V/J contribution is a per-position conservation profile $w(p) = \text{Pr}[\text{position } p \text{ is germline-retained}]$, determined by the gene’s trimming-length distribution, rather than a substitution matrix learned from data. Combined, the NDN matrix scores the core and $w(p)$ down-weights substitutions at germline-retained flanks (realized through a per-position weighting of the base matrix). The germline-retention profile $w(p)$ is computed from the OLGA recombination model (per-gene V/J deletion distributions and germline CDR3-portion segment lengths, via `mirpy’s mir.basic.trimming`) and bundled with `vdjmatch`; Fig. 2 shows it drops monotonically from each anchor (the CASS V-stem and the J motif are germline; the core is insert). Region-aware scoring weights each CDR3 column by $1 - \text{Pr}[\text{germline}]$ and applies the NDN matrix — implemented in `vdjmatch.match.regions` and evaluated below.

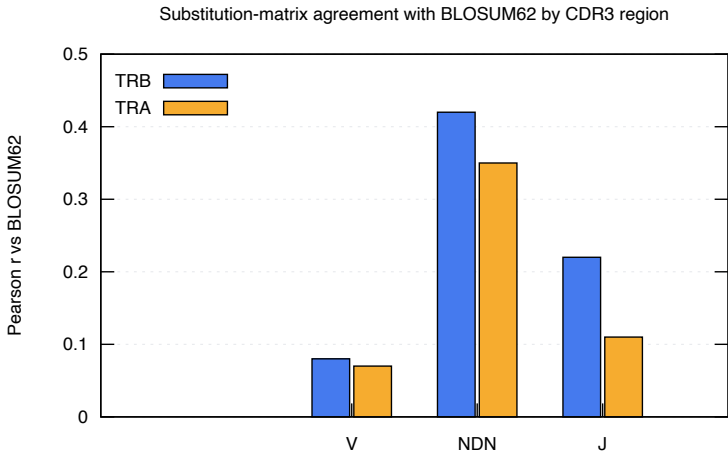


Figure 1: Per-region agreement of the learned CDR3 substitution matrix with BLOSUM62 (Pearson r of off-diagonal scores), human VDJdb. The free-substitution signal lives in the NDN core; the V flank is germline-determined.

3. SCORE AGGREGATION AND SIGNIFICANCE

PER-HIT TO PER-QUERY.. A region-aware similarity score sums the NDN matrix score over the core and the (down-weighted) flank contributions; this is available as an optional ranking score and

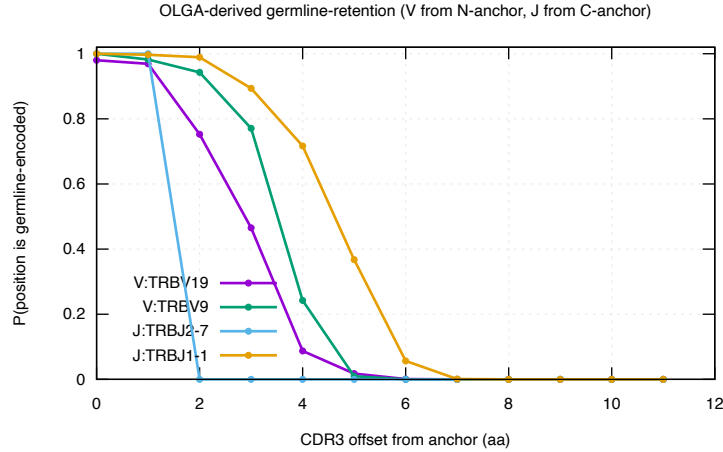


Figure 2: OLGA-derived germline-retention $w(p)$ (via `mirpy`): probability that each CDR3 offset from the V-Cys (or J) anchor is germline-encoded. It falls monotonically into the insert core, defining the region weight $1 - w(p)$ used in scoring.

generalizes the legacy cloglog generalized-linear aggregation over V/CDR1/CDR2/J/CDR3.

CONTROL-CALIBRATED E-VALUE (PRIMARY).. Immune repertoires are biologically redundant, so an i.i.d. null over-calls. `vdjmatch` instead counts a query’s VDJdb neighbours within the ball and compares to the count expected from a matched background control: with $N = |D|$, M the control size and n_C the control neighbour count, $\mathbb{E} = (N/M) n_C$ is the expected number of VDJdb neighbours by chance and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D)$ is the enrichment significance — significant only when a clonotype has *more* VDJdb neighbours than the generative background predicts. The Poisson/Chen–Stein theory, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix.

PAIRED $\alpha\beta$ E-VALUE.. The null factorization is not an approximation of convenience: α and β chains pair *almost without constraint* in the repertoire at large, as shown by integrating structural and paired-sequencing data [1], so under the background (generative) law the joint ball mass genuinely factorizes, $\pi_0^{\alpha\beta} \approx \pi_0^\alpha \pi_0^\beta$. Crucially, the *same* analysis finds that *antigen-driven selection distorts* this uniform pairing landscape — i.e. conditioned on an epitope the two chains are *not* independent. This is exactly the structure the joint E-value exploits: the null factorizes (background pairing is unconstrained), so among the N paired VDJdb complexes the expected number of chance joint matches is $\mathbb{E}_{\alpha\beta} = N \pi_0^\alpha \pi_0^\beta$ with p the Poisson tail on the count of complexes matched in *both* chains; an excess over this factorized null is precisely the epitope-conditional dependence (co-selection of α and β) that signals a true specificity. Because the joint null is far smaller than either chain’s, true pairs are dramatically more significant: on the paired TCRvdb benchmark, 296/300 pairs match a complex and 295/296 (99.7%) recover the correct epitope, with joint p down to 3.6×10^{-9} . (When the two chains’ hits are treated as independent tests, the equivalent statistic is Fisher’s combination of the per-chain enrichments; the appendix `seqtree` co-occupancy term quantifies the residual dependence.)

4. RESULTS: DOES VDJAM (AND REGION WEIGHTING) HELP?

A matrix derived from same-antigen pairs must be tested on epitopes it never saw, or the evaluation is circular. For each held-out epitope E^* we re-derive the NDN matrix from *all other* epitopes, fix the candidate set with a single unit-cost search on E^* (exact self excluded), and re-score each (query, hit) pair four ways — unit, BLOSUM62, NDN-VDJAM, and region-weighted NDN-VDJAM — ranking by score and reporting micro PR-AUC for “hit shares E^* ” (Fig. 3, eight largest TRB epitopes). Two honest findings. (i) *Region weighting is a real, consistent, but small improvement over the flat matrix* (+0.004 at scope 2, +0.007 at scope 3; never worse on any epitope, and the gain grows with scope) — modest because CDR3 variation is *already* NDN-concentrated, so down-weighting the rarely-varying germline flanks moves few hits; it should matter more for cross-V-gene comparisons, where flank (germline-allele) differences are common. (ii) *BLOSUM62 remains the strongest substitution matrix at this narrow-scope retrieval task* (mean PR-AUC 0.32 vs 0.28 region-VDJAM vs 0.29 unit): the data-derived NDN matrix captures general substitution tolerance that BLOSUM already encodes well, and with limited same-antigen pairs it is noisier. At scope 2 the search ball supplies most of the specific-vs-nonspecific discrimination and any matrix only re-ranks within it; the substitution model is therefore a second-order lever here, and the first-order signal is the control-calibrated neighbour count (§3).

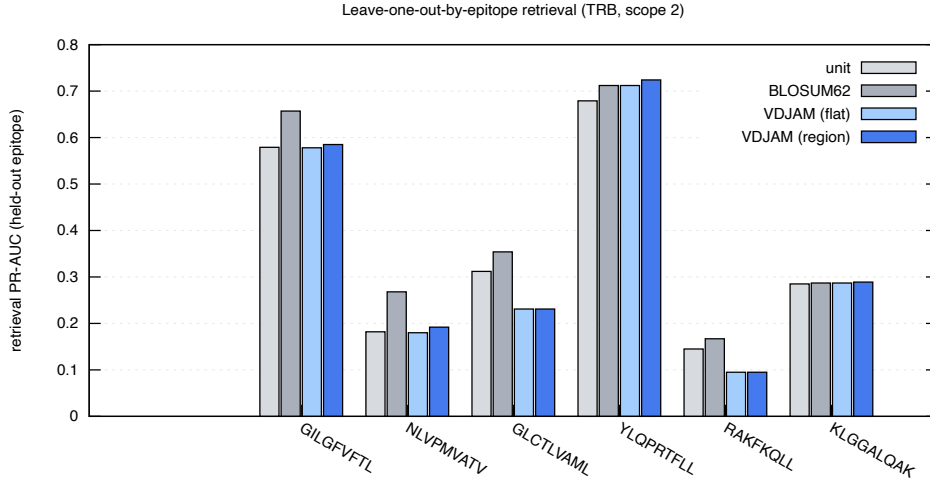


Figure 3: Leave-one-out-by-epitope retrieval PR-AUC (TRB, scope 2): the NDN VDJAM matrix versus BLOSUM62 and unit cost on epitopes excluded from matrix estimation.

5. WORKED EXAMPLES

PAIRWISE (DIFFERENT LENGTHS).. seqtree aligns CDR3s of differing length, placing indels where the loop genuinely lengthens. The edit-distance alignments below (influenza GIL-specific TRB CDR3s) show the conserved CASS V-anchor and DTQYF J-motif with substitutions and indels confined to the central NDN core:

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query (ref): CASSPRSGETQYF    epitope GILGFVFTL    (unit / edit-distance alignment)

vs CASSNRSGETQYF (len 13; subs=1 ins=0 dels=0; CIGAR 4=1X8=)
  CASSPRSGETQYF
  |||| |||||
  CASSNRSGETQYF

vs CASSIRSGESTQYF (len 14; subs=1 ins=1 dels=0; CIGAR 4=1X4=1I4=)
  CASSPRSGE-TQYF
  |||| |||| |||
  CASSIRSGESTQYF

vs CASSLFSETQYF (len 12; subs=2 ins=0 dels=1; CIGAR 4=2X1=1D5=)
  CASSPRSGETQYF
  |||| | ||||
  CASSLFS-ETQYF

vs CASSLGGETQYF (len 11; subs=1 ins=0 dels=2; CIGAR 4=2D1X6=)
  CASSPRSGETQYF
  ||||  |||||
  CASS--LGETQYF

vs CASSPWSGSQETQYF (len 15; subs=1 ins=2 dels=0; CIGAR 5=1X2=2I5=)
  CASSPRSG--ETQYF
  ||||| ||  ||||
  CASSPWSGSQETQYF

```

MULTIPLE (SAME LENGTH).. Aligning same-length same-epitope CDR3s (no indels needed) makes the conservation profile explicit (Figs. 4–5): columns are shaded by conservation, dark at the germline-determined CASS prefix and J motif, light through the variable NDN core. This is the structural counterpart of the finding of Shcherbinin et al. [2] that specificity-preserving CDR3 substitutions occur predominantly outside the peptide/MHC-contacting positions while preserving the loop backbone — i.e. the tolerated variation we score in the NDN core is exactly the variation that does not break recognition.

6. LIMITATIONS AND ROADMAP

The NDN matrix is estimated from a limited set of same-antigen single-substitution pairs and, at narrow scope, does not beat BLOSUM62; the region-aware germline+trimming weighting (now implemented) gives a small, consistent, scope-growing gain over the flat matrix but not over BLOSUM62 at this task. The clearest remaining lever is the *V-gene* dimension: a V-gene similarity term (grouping V genes by CDR1/CDR2/CDR2.5 distance) and cross-V comparisons, where germline-flank differences are common and the region weighting should matter most; a per-chain treatment (TRA vs TRB) is also pending. Separately, the control’s V–J usage must match the query repertoire’s or the

GILGFVFTL (TRB CDR3, length 13, n=14)

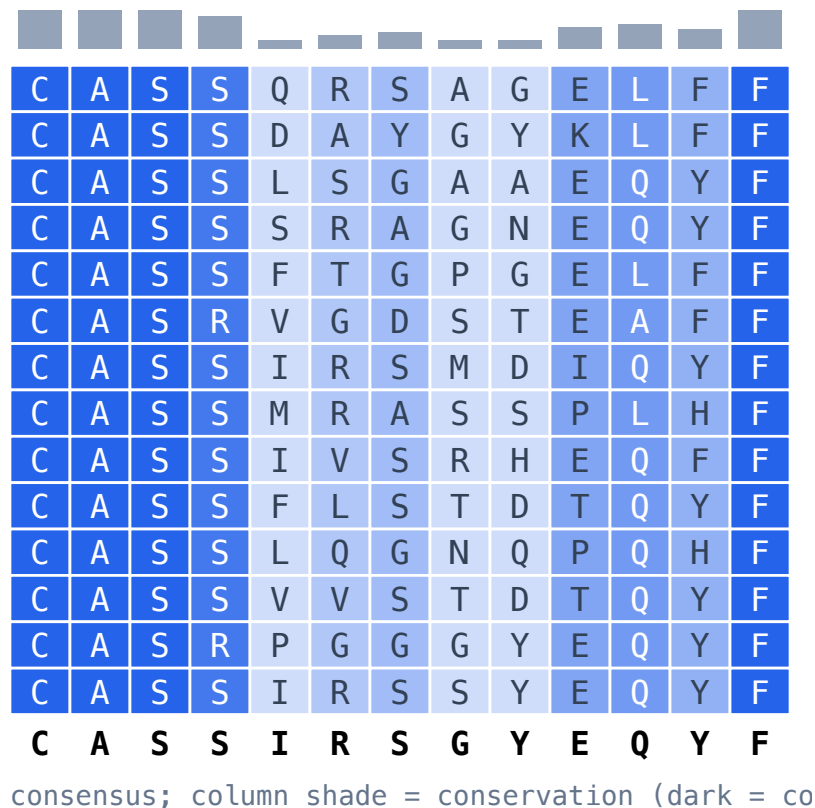


Figure 4: GIL (influenza M1, A*02) TRB CDR3 block; conserved V/J flanks, variable NDN core.

null is mis-specified — whether to renormalize V–J usage (at the risk of erasing genuine V-gene bias) is an open question to settle on held-out epitopes. These extensions, and the full benchmark against existing tools, are ongoing work.

REFERENCES

- [1] Dmitrii S. Shcherbinin, Vlad A. Belousov, and Mikhail Shugay. Comprehensive analysis of structural and sequencing data reveals almost unconstrained chain pairing in TCR $\alpha\beta$ complex. *PLoS Computational Biology*, 16(3):e1007714, 2020.
- [2] Dmitrii S. Shcherbinin, Vadim K. Karnaukhov, Ivan V. Zvyagin, Dmitriy M. Chudakov, and Mikhail Shugay. Large-scale template-based structural modeling of T-cell receptors with known antigen specificity reveals complementarity features. *Frontiers in Immunology*, 14:1224969, 2023.

NLVPMVATV (TRB CDR3, length 15, n=14)

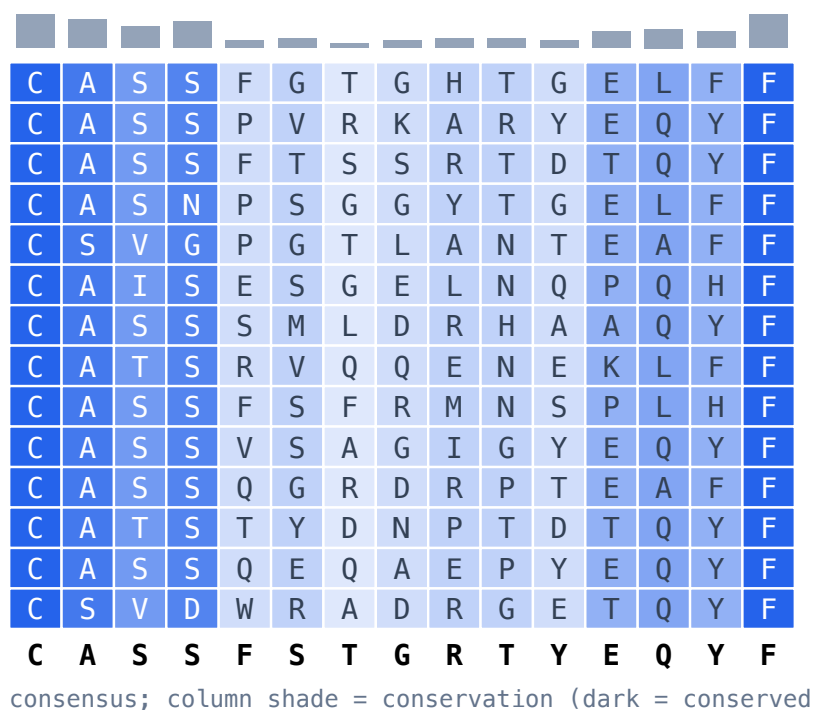


Figure 5: NLV (CMV pp65, A*02) TRB CDR3 block; same conservation pattern.